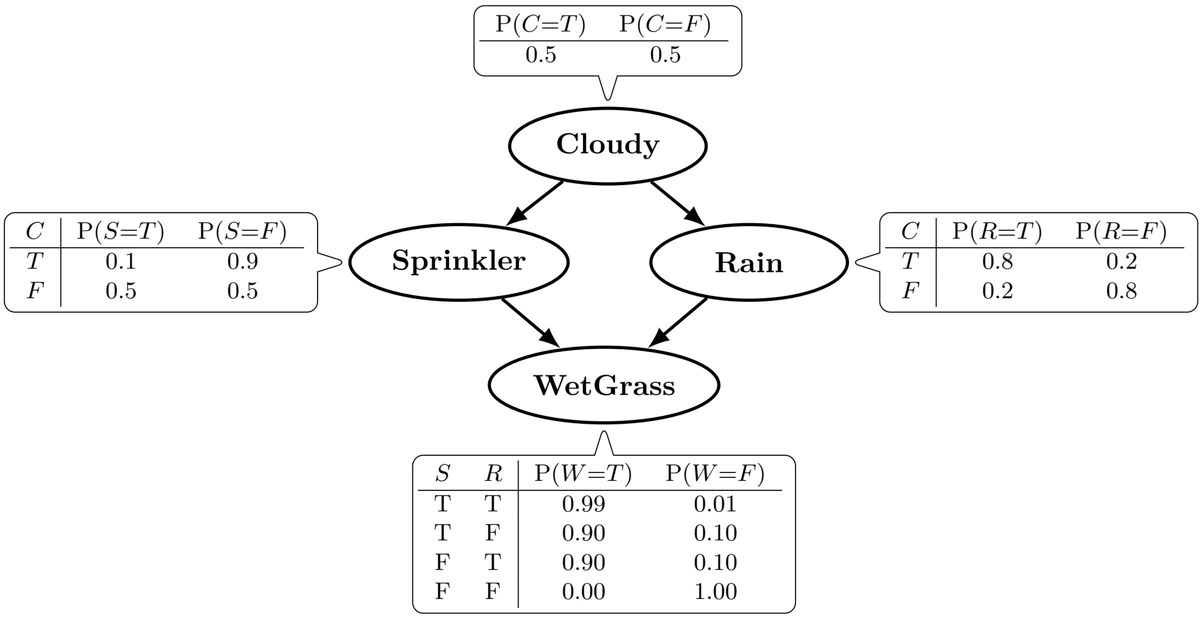
9. Using the graph below, please answer the questions and explain how to apply Bayesian inference; identify the conditional probabilities from the graph, and show each step, including total probability and any conditional dependencies.



P(C=T)	P(C=F)
0.5	0.5

C	P(S=T)	P(S=F)
T	0.1	0.9
F	0.5	0.5

C	P(R=T)	P(R=F)
T	0.8	0.2
F	0.2	0.8

S	R	P(W=T)	P(W=F)
T	T	0.99	0.01
T	F	0.90	0.10
F	T	0.90	0.10
F	F	0.00	1.00

C = Cloudy, S = Sprinkler, R = Rain, W = WetGrass, T = True, F = False.

a. (1pt) Calculate the marginal probability that the grass is wet.

Answer: Factorization (from the DAG and conditional independence $S \perp R \mid C$):

$$P(C, S, R, W) = P(C)\, P(S \mid C)\, P(R \mid C)\, P(W \mid S, R)$$

CPTs:

$$\begin{aligned}
&P(C{=}T) = P(C{=}F) = 0.5,\\
&P(R{=}T \mid C{=}T) = 0.8, \quad P(R{=}T \mid C{=}F) = 0.2,\\
&P(S{=}T \mid C{=}T) = 0.1, \quad P(S{=}T \mid C{=}F) = 0.5,\\
&P(W{=}T \mid S{=}T, R{=}T) = 0.99, \quad P(W{=}T \mid S{=}T, R{=}F) = 0.9,\\
&P(W{=}T \mid S{=}F, R{=}T) = 0.9, \quad P(W{=}T \mid S{=}F, R{=}F) = 0.
\end{aligned}$$

$$P(W = T) = \sum_{c \in \{T,F\}} \sum_{s \in \{T,F\}} \sum_{r \in \{T,F\}} P(C = c)\, P(S = s \mid C = c)\, P(R = r \mid C = c)\, P(W = T \mid S = s, R = r)$$

$$\begin{aligned}
P(W = T) = &\underbrace{0.5 \cdot 0.1 \cdot 0.8 \cdot 0.99}_{C=T,S=T,R=T} + \underbrace{0.5 \cdot 0.1 \cdot 0.2 \cdot 0.90}_{C=T,S=T,R=F}\\
&+ \underbrace{0.5 \cdot 0.9 \cdot 0.8 \cdot 0.90}_{C=T,S=F,R=T} + \underbrace{0.5 \cdot 0.9 \cdot 0.2 \cdot 0.00}_{C=T,S=F,R=F}\\
&+ \underbrace{0.5 \cdot 0.5 \cdot 0.2 \cdot 0.99}_{C=F,S=T,R=T} + \underbrace{0.5 \cdot 0.5 \cdot 0.8 \cdot 0.90}_{C=F,S=T,R=F}\\
&+ \underbrace{0.5 \cdot 0.5 \cdot 0.2 \cdot 0.90}_{C=F,S=F,R=T} + \underbrace{0.5 \cdot 0.5 \cdot 0.8 \cdot 0.00}_{C=F,S=F,R=F}
\end{aligned}$$

$$\begin{aligned}
&= 0.0396 + 0.0090 + 0.3240 + 0.0000\\
&\quad + 0.0495 + 0.1800 + 0.0450 + 0.0000 = 0.6471
\end{aligned}$$

$$\boxed{P(W = T) \approx 0.6471}$$

b. (1pt) Compute $P(\text{Cloudy} = \text{True} \mid \text{Sprinkler} = \text{True})$

Answer:

$$P(C = T \mid S = T) \; = \; \frac{P(S = T \mid C = T)\, P(C = T)}{P(S = T)}$$

$$\text{Denominator: } P(S = T) \; = \; P(S = T \mid C = T)\, P(C = T) + P(S = T \mid C = F)\, P(C = F)$$

$$P(S = T) \; = \; (0.1)(0.5) + (0.5)(0.5) = 0.05 + 0.25 = 0.30$$

$$\text{Numerator: } P(S = T \mid C = T)\, P(C = T) = (0.1)(0.5) = 0.05$$

$$P(C = T \mid S = T) = \frac{0.05}{0.30} = \frac{1}{6} \approx 0.167$$

$$\boxed{P(C = T \mid S = T) \approx 0.167}$$

c. (1pt) Given that the grass is wet, compute the posterior probability that it is cloudy.

Answer:

$$P(C = T \mid W = T) \; = \; \frac{P(W = T \mid C = T)\, P(C = T)}{P(W = T)}$$

$$P(W = T \mid C = T) \; = \; \sum_{s \in \{T,F\}} \sum_{r \in \{T,F\}} P(S = s \mid C = T)\, P(R = r \mid C = T)\, P(W = T \mid S = s, R = r)$$

$$\begin{aligned} P(W = T \mid C = T) &= (0.1)(0.8)(0.99) + (0.1)(0.2)(0.90) \\ &\quad + (0.9)(0.8)(0.90) + (0.9)(0.2)(0.00) \\ &= 0.0792 + 0.0180 + 0.6480 + 0.0000 = 0.7452 \end{aligned}$$

$$P(W = T) = 0.6471 \quad \text{(computed earlier)}$$

$$P(C = T \mid W = T) = \frac{(0.7452)(0.5)}{0.6471} = \frac{0.3726}{0.6471} \approx 0.576$$

$$\boxed{P(C = T \mid W = T) \approx 0.576}$$

$$P(C = F \mid W = T) \; = \; \frac{P(W = T \mid C = F)\, P(C = F)}{P(W = T)}$$

$$P(W = T \mid C = F) \; = \; \sum_{s \in \{T,F\}} \sum_{r \in \{T,F\}} P(S = s \mid C = F)\, P(R = r \mid C = F)\, P(W = T \mid S = s, R = r)$$

$$\begin{aligned} P(W = T \mid C = F) &= (0.5)(0.2)(0.99) + (0.5)(0.8)(0.90) \\ &\quad + (0.5)(0.2)(0.90) + (0.5)(0.8)(0.00) \\ &= 0.0990 + 0.3600 + 0.0900 + 0.0000 = 0.5490 \end{aligned}$$

$$P(W = T) = 0.6471 \quad \text{(computed earlier)}$$

$$\begin{aligned} P(C = F \mid W = T) &= \frac{(0.5490)(0.5)}{0.6471} \\ &= \frac{0.2745}{0.6471} \\ &\approx 0.424 \end{aligned}$$

$$\boxed{P(C = F \mid W = T) \approx 0.424}$$

$$\text{Check: } P(C = T \mid W = T) + P(C = F \mid W = T) = 0.576 + 0.424 = 1.000$$